

# Dietary Supplement Research: Evidence-Based Summary

**Disclaimer:** This document is for informational purposes only and does not constitute medical advice. Individuals with medical conditions or taking medications should consult a healthcare professional before beginning any new supplement.

| Supplement                 | Recommended Dosage                                                                            | Evidence Strength                                                     | Starting Age                                                                                     |
|----------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Creatine (mono)</b>     | 20 g/day (4×5 g) loading for 5–7 d, then 3–5 g/day maintenance 【10†L268-L277】 【10†L278-L283】  | Strong (improves strength) 【7†L79-L87】                                | Teens & up (studies include adolescents 【20†L901-L909】 )                                         |
| <b>Whey/Casein Protein</b> | 20–40 g per serving (≈0.25–0.4 g/kg) 【36†L94-L102】 , spread every 3–4 h 【36†L102-L110】        | Strong (supports muscle mass) 【36†L84-L92】 【36†L94-L102】              | From ~9–13 y (adolescent RDA) and up; older adults often need ≥ protein to preserve muscle       |
| <b>Omega-3 (EPA+DHA)</b>   | ~250–500 mg EPA+DHA daily (1 g/day for heart disease risk 【40†L33-L39】 ; 2–4 g for high TG)   | Moderate (lowers TG, CVD prevention) 【40†L33-L39】 【42†L91-L94】        | From infancy (with careful dosing) to seniors; pregnancy (DHA)                                   |
| <b>Vitamin D3 (+K2)</b>    | 800–2000 IU/day (20–50 µg); often ↑ dose in winter or high-risk 【44†L251-L254】 【44†L263-L270】 | Strong (bone & muscle health) 【44†L253-L261】 【44†L263-L270】           | All ages (infants 400 IU, children ≥600 IU 【44†L229-L238】 ; seniors 800–1000 IU 【44†L251-L254】 ) |
| <b>Magnesium</b>           | 200–400 mg/day (as citrate or glycinate) 【50†L229-L233】                                       | Moderate (corrects deficiency; minor exercise benefit) 【50†L148-L157】 | From childhood (RDA 80–410 mg by age 【50†L229-L233】 ) and up; seniors often need supplement      |
| <b>Zinc</b>                | ~8–11 mg/day (RDA); supplements 15–30 mg if deficient 【57†L226-L229】 【60†L24-L29】             | Moderate (immune & wound healing) 【60†L1-L4】                          | From childhood (RDA 5–11 mg) and up; pregnant women 12–13 mg 【57†L226-L229】                      |
| <b>Vitamin B12</b>         | ~2.4 µg/day (RDA); vegans/ seniors often supplement 50–500 µg                                 | Strong (prevents anemia/neuro issues) 【62†L237-L245】 【64†L60-L64】     | All ages (infants AI ~0.4–0.5 µg, children RDA ~0.9–1.8 µg 【62†L239-L245】 )                      |

| Supplement                  | Recommended Dosage                                                          | Evidence Strength                                                                      | Starting Age                                                                              |
|-----------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Iron</b>                 | 8–18 mg/day (men 8 mg; women 18 mg; 27 mg in pregnancy)                     | Strong (prevents anemia; performance)                                                  | Children from ~6 mo (1 mg/kg), adolescents & menstruating women (↑), seniors RDA declines |
| <b>Multivitamin/Mineral</b> | As label (often 1 tablet/day covering ~100% DV)                             | Weak (no clear performance benefit) <b>【71†L340-L347】</b>                              | Generally ≥2 y (chewable for kids); special formulas for seniors                          |
| <b>Beta-Alanine</b>         | 3.2–6.4 g/day (split doses) <b>【75†L2-L4】</b>                               | Moderate (improves 1–4 min performance)                                                | Adults (studies in young adults; unknown in minors)                                       |
| <b>Caffeine</b>             | 3–6 mg/kg (~200–400 mg) 30–60 min pre-exercise <b>【84†L65-L74】</b>          | Strong (enhances endurance & power) <b>【84†L65-L74】</b>                                | Adults and older teens (avoid high doses in children)                                     |
| <b>Citrulline Malate</b>    | ~6–8 g 1 h pre-exercise (2:1 citrulline:malate)                             | Weak-to-moderate (may boost endurance)                                                 | Adults (limited data in minors)                                                           |
| <b>Ashwagandha</b>          | 300–600 mg/day (standardized extract)                                       | Moderate (reduces stress; may ↑ testosterone)                                          | Adults; avoid in pregnancy (traditionally a uterine stimulant)                            |
| <b>Collagen</b>             | 2.5–15 g/day (hydrolyzed peptides) or 40 mg/day UC-II <b>【95†L280-L286】</b> | Moderate (improves skin, joint comfort) <b>【95†L241-L249】</b><br><b>【95†L262-L264】</b> | Adults (source-dependent; not for vegans)                                                 |
| <b>Probiotics</b>           | 1×10 <sup>9</sup> –10 <sup>11</sup> CFU/day (strain-specific)               | Moderate (gut/immune health; may aid endurance) <b>【98†L362-L371】</b>                  | All ages (infant formulas on label; immune benefits)                                      |
| <b>Electrolytes</b>         | Replace losses: ~300–700 mg NaCl per L of sweat; K, Mg as needed            | Moderate (prevents dehydration/cramps)                                                 | All ages (sports drinks for children in extreme exercise with caution)                    |
| <b>Glutamine</b>            | 5–10 g/day                                                                  | Weak (no clear benefit for performance in healthy people)                              | Adults (conditionally essential during illness; generally safe)                           |
| <b>BCAAs/EAA</b> s          | 5–10 g BCAAs or 10–15 g EAAs                                                | Weak (useless if protein intake adequate)                                              | Adults (children/teens get EAAs from diet)                                                |

## Creatine (Monohydrate)

- **Mechanism:** Creatine is stored in muscle as phosphocreatine (PCr) and donates its phosphate to regenerate ATP during short, intense exercise. This buffering of energy helps sustain high-power output [22†L153-L161] [99†L160-L168] .
- **Dosage:** A common regimen is a loading phase of 20 g/day (4×5 g) for 5–7 days, then a maintenance dose of 3–5 g/day [10†L268-L277] [10†L278-L283] . Larger individuals or vegetarians (with lower baseline levels [99†L195-L203] ) may require up to 5–10 g maintenance. Taking creatine with carbohydrates (or a carb+protein mix) can enhance uptake [10†L281-L288] .
- **Timing:** Timing is flexible – studies show taking creatine before or after training leads to similar gains. Many prefer post-workout or with meals to boost uptake, though daily consistency is key [10†L268-L277] [10†L278-L283] .
- **Benefits:** Creatine has *strong* evidence for improving muscle strength, power, and high-intensity performance [7†L79-L87] . It reliably increases lean mass and training adaptations when combined with resistance exercise [7†L79-L87] . Additional benefits include faster recovery and potential cognitive gains (especially in sleep-deprived states). Vegetarians/vegans often see larger effects due to low dietary intake [99†L195-L203] .
- **Side Effects:** In healthy people, creatine is extremely safe. The only consistent effect is a slight weight gain (water retention) [20†L833-L841] . Large reviews report no adverse impacts on kidney or liver function even long-term [20†L883-L892] . Mild GI discomfort or cramps are rare [20†L833-L841] . In those with kidney disease, caution is advised since creatine is metabolized and excreted via the kidneys.
- **Age:** Studies show benefits and safety in adolescents and older adults [20†L901-L909] [20†L913-L921] . ISSN notes no harm seen in teens at usual doses, though most research is in 18+ [20†L901-L909] . Children under puberty generally don't need it; use in minors should be considered case-by-case with medical advice.
- **Contraindications:** Avoid if you have pre-existing kidney issues (or always check with a doctor). Also use caution in pregnancy or breastfeeding (insufficient data).
- **Target Audience:** Especially beneficial for strength/power athletes, bodybuilders, and older adults to preserve muscle mass. Vegetarians/vegans often take creatine to make up for low dietary creatine. It may also help anyone doing repeated sprints or heavy lifts.
- **Quality:** Use pure **creatine monohydrate**, preferably micronized for mixability. Look for third-party certification (e.g. NSF Certified for Sport, Informed-Choice) to ensure purity (creatine is cheap so extra fillers are unnecessary).
- **Cost-Benefit:** Creatine is very inexpensive per serving and well-supported by research, so the cost is low relative to its strong evidence of benefit [20†L833-L841] .

## Whey/Casein Protein

- **Mechanism:** Whey and casein are milk proteins rich in essential amino acids (especially leucine) that stimulate muscle protein synthesis. Whey is rapidly absorbed, spiking blood amino acids; casein digests slowly, providing a steady amino acid supply (often used overnight) [36†L94-L102] [36†L131-L134] .
- **Dosage:** The ISSN recommends 1.4–2.0 g protein/kg/day for active individuals [36†L84-L92] . In practice, 20–40 g high-quality protein per meal (or ~0.25–0.4 g/kg) maximizes muscle-building

【36†L94-L102】. Around 700–3,000 mg of leucine per dose is ideal. Casein-specific: ~30–40 g before bed boosts overnight recovery 【36†L131-L134】.

- **Timing:** Protein can be taken anytime; spreading doses evenly (every 3–4 h) is optimal 【36†L102-L110】. Consuming whey immediately after training synergizes with exercise. Casein taken pre-sleep (30–40 g) enhances overnight muscle synthesis 【36†L131-L134】. Unlike past “anabolic window” ideas, total daily intake matters more than exact timing.
- **Benefits:** Strong evidence shows protein supplements help build and maintain muscle when used with resistance training 【36†L84-L92】. Whey in particular has been linked to greater gains in lean mass and strength versus training alone 【36†L84-L92】 【22†L134-L142】. Protein also aids recovery. In groups like the elderly or those with inadequate dietary protein, supplementation helps prevent frailty.
- **Side Effects:** Generally well-tolerated. Some people may experience bloating or gas, especially with whey concentrate (lactose). Those with milk allergies should avoid dairy proteins or use plant proteins. In individuals with kidney disease, very high protein intake can be a concern (but normal supplemental doses are safe in healthy people).
- **Age:** Safe from childhood (with adjusted doses); many pediatric formulas are whey-based. Teen and adult athletes benefit most. Older adults often require higher protein to counter muscle loss.
- **Contraindications:** Dairy allergy or lactose intolerance (unless using isolate). Kidney disease patients should talk to a doctor about high-protein diets.
- **Target Audience:** Athletes, bodybuilders, or anyone wanting to ensure adequate protein (e.g. those who struggle to meet protein needs with food alone). Vegans might use plant-based protein instead; vegetarians usually get enough via dairy.
- **Quality:** Choose proteins with high protein percentage and minimal additives. Look for NSF, Informed-Sport, or USP-verified brands to avoid contamination and verify amino acid content.
- **Cost-Benefit:** Protein powders range from moderate to high cost. Whey concentrate is cheapest; isolates/isoflavones cost more. Given the solid evidence for muscle benefits, cost is generally justified for serious trainers. Vegetarians/vegans may find plant proteins useful but verify complete amino acid profiles.

## Omega-3 Fatty Acids (EPA/DHA)

- **Mechanism:** Long-chain omega-3s (EPA and DHA) from fish or algae oils incorporate into cell membranes and serve as precursors to anti-inflammatory eicosanoids. They reduce inflammation and triglycerides, and may improve endothelial function.
- **Dosage:** For general health, 250–500 mg combined EPA+DHA/day is often recommended. The AHA suggests ~1 g/day for people with coronary disease 【40†L33-L39】. Higher doses (2–4 g/day) are used under medical supervision to lower very high triglycerides 【42†L91-L94】. Supplements usually contain ~180 mg EPA + 120 mg DHA per 1 g capsule 【38†L376-L382】.
- **Timing:** Fat-soluble vitamins – take with a meal containing fat to enhance absorption 【44†L276-L280】. No specific pre/post exercise timing needed.
- **Benefits:** Evidence is **strong** that omega-3s lower blood triglycerides dose-dependently (about – 6 mg/dL per gram 【42†L91-L94】). There is moderate evidence for cardiovascular benefits (e.g. reduced risk of heart events) at higher intakes 【40†L33-L39】. Some studies show modest improvement in joint pain and cognitive decline. Athletes may find mild benefits in reducing exercise-induced inflammation or aiding recovery, but performance effects are modest. Omega-3s also support brain and eye health, especially during pregnancy (DHA).

- **Side Effects:** Generally safe. High doses (>3–4 g/day) can increase bleeding risk (platelet aggregation) and cause gastrointestinal upset (fishy burp). Very high vitamin A (from cod liver oil) can be toxic; so stick to purified oils. Fish allergies are not triggered by purified omega-3 (no proteins remain). Those on blood thinners should consult a doctor.
- **Age:** Essential for all ages. Infants and children benefit from DHA (found in breastmilk or fortified formula). Older adults often have lower dietary intake and can supplement safely. Pregnant and nursing women are encouraged to get DHA for fetal/neonatal brain development.
- **Contraindications:** People with bleeding disorders or on anticoagulants should use caution at high doses. Fish oil supplements often remove mercury, so purity matters.
- **Target Audience:** Individuals with little fish in their diet (vegans should use algae-based DHA), those with high triglycerides, or with inflammatory conditions. Some athletes use fish oil hoping to speed recovery, though evidence here is limited.
- **Quality:** Choose products certified for purity (e.g. IFOS rating, NSF, Informed-Choice). Ensure low peroxide values (i.e. not rancid). Algae-based DHA is a vegetarian alternative. Check labels to avoid “proprietary blends” – doses of EPA/DHA should be clear.
- **Cost-Benefit:** Fish oil is relatively affordable and well-researched. Given the cardiovascular benefits and safety, moderate dosing is cost-effective. Algal oil is pricier but provides DHA without fish.

## Vitamin D3 (+K2)

- **Mechanism:** Vitamin D3 (cholecalciferol) is converted in the liver/kidneys to active calcitriol, which enhances calcium absorption in the gut and regulates bone metabolism. Vitamin K2 (menaquinone) activates proteins (like osteocalcin) that incorporate calcium into bone and help prevent vascular calcification. Together, D3 and K2 promote healthy bone mineralization.
- **Dosage:** RDA is ~600–800 IU (15–20 µg) for adults, but many experts recommend higher for adequacy. Elderly and those at risk often take 800–1000 IU/day [44†L251-L254] . For deficiency correction, 2000–4000 IU/day is common (with monitoring). Vitamin K2 (often as MK-7) is typically 90–200 µg/day, though no official RDA exists.
- **Timing:** Take D3 with a fat-containing meal (better absorption [44†L276-L280] ). K2 can be taken with or without food. There’s no specific “time of day” requirement, but avoid extreme single mega-doses (yearly high doses have been linked to falls [44†L268-L274] ).
- **Benefits:** Strong evidence supports vitamin D for bone health. Supplements (usually with calcium) can increase bone density and reduce fall/fall risk in deficient individuals [44†L253-L261] [44†L263-L270] . D also has roles in muscle function, immunity, and mood, though those benefits are less certain. K2’s evidence is emerging; combined D+K appears synergistic for bones (fewer fractures) [44†L253-L261] and may protect against arterial calcification in theory.
- **Side Effects:** Safe at recommended doses. Excessive vitamin D (>50,000 IU/day) can cause hypercalcemia, leading to nausea, weakness, kidney stones. K2 has no known toxicity at usual supplement doses, but high K2 could interact with blood thinners (vitamin K antagonizes warfarin).
- **Age:** All ages need vitamin D for bone growth/maintenance. Infants (starting ~400 IU in formula) and children should meet RDA (600 IU). Teens and adults should get 600–800 IU. Seniors (60+ years) often need at least 800–1000 IU to maintain 25(OH)D >50 nmol/L [44†L251-L254] .
- **Contraindications:** Conditions of hypercalcemia (e.g. some cancers, sarcoidosis) contraindicate high D. Pregnancy and breastfeeding: moderate supplementation (600–1000 IU) is safe and often recommended (D deficiency in mothers can harm infant). K2 in pregnancy has been less studied—usually included in prenatal vitamins in modest amounts.

- **Target Audience:** People in northern latitudes or who have limited sun exposure, the elderly (skin makes less D3) [44†L179-L187] , dark-skinned individuals, or those with osteoporosis/osteopenia. K2 is often paired with D in bone-targeting supplements.
- **Quality:** Use D3 (cholecalciferol), as it's more potent than D2 [44†L281-L284] . Supplements often list "IU" of D3 directly. For K2, look for "MK-7" or "MK-4" forms; MK-7 has longer half-life. Choose supplements tested for purity and correct dosing.
- **Cost-Benefit:** Vitamin D is inexpensive per dose. Because deficiency is common (especially in winter), it's often cost-effective to supplement. K2 is pricier; evidence is less established than D3 alone, but many experts consider combined D+K a reasonable strategy for bone health.

## Magnesium

- **Mechanism:** Magnesium is a cofactor in >300 enzymatic reactions – crucial for ATP production, muscle contraction/relaxation, nerve transmission, and bone structure [50†L148-L157] . It helps regulate calcium and potassium transport across cell membranes, affecting muscle and nerve function [50†L153-L157] .
- **Dosage:** RDAs are 400–420 mg/day for adult men and 310–320 mg/day for adult women [50†L229-L233] . Many supplements provide 100–400 mg of elemental magnesium (commonly as citrate, glycinate, malate, etc.) per tablet or dose. If correcting deficiency, doses up to 400 mg/day can be used (note: the UL for supplemental magnesium is 350 mg to avoid diarrhea [52†L703-L710] ).
- **Timing:** Can be taken anytime; some prefer evening (magnesium can have a mild calming effect). If using citrate/oxide forms, splitting doses reduces laxative effect. With meals is fine, although food can slightly reduce absorption.
- **Benefits:** Correcting suboptimal magnesium can improve muscle and nerve function. Evidence for athletic performance is **moderate/unclear** – it's not a direct "ergogenic aid," but magnesium deficiency (common in athletes and elderly) can impair performance and cause cramps. Supplements may relieve muscle cramps or improve sleep quality in those who are deficient. Otherwise, benefits are mostly seen in deficiency states.
- **Side Effects:** High doses (especially Mg oxide or citrate) can cause loose stools or diarrhea. The tolerable upper intake level (supplements only) is 350 mg/day for adults [52†L703-L710] . Above that, diarrhea is likely. People with normal kidney function excrete excess magnesium; those with kidney failure can accumulate it, risking toxicity (muscle weakness, arrhythmias).
- **Age:** Children and teens have lower RDAs (see RDA table [50†L225-L233] ). The elderly often have lower magnesium due to reduced intake/absorption and prescription medications, so supplementation is common. All ages need some magnesium; severe deficiency in kids is rare.
- **Contraindications:** Severe renal impairment (risk of hypermagnesemia). Use caution if taking medications: e.g. magnesium can bind to certain antibiotics (tetracyclines, quinolones) and bisphosphonates [52†L731-L739] , so space dosing 2–4 hours apart. Proton pump inhibitors (PPIs) can cause magnesium depletion over long-term [52†L748-L756] .
- **Target Audience:** Anyone with poor diet (since many people don't get RDA from food), athletes (sweating loses Mg), diabetics or those on diuretics (can cause magnesium loss). Also those with muscle cramps or poor sleep.
- **Quality:** Forms like citrate, glycinate or malate are better absorbed than oxide [52†L703-L710] . A label will list elemental magnesium. Look for brands tested by USP or NSF to ensure label accuracy. Avoid excessive "elemental" concentration at once.
- **Cost-Benefit:** Magnesium is very affordable. Given its role in many body functions, ensuring adequate intake is worthwhile. For performance, benefit is modest unless correcting a shortfall.

## Zinc

- **Mechanism:** Zinc is an essential trace mineral involved in immune function, DNA/RNA synthesis, wound healing, and protein metabolism 【60†L1-L4】 . It acts as a cofactor for hundreds of enzymes and is crucial for healthy growth and taste perception.
- **Dosage:** RDA is 11 mg/day for adult men, 8 mg/day for women (ages  $\geq 19$ ) 【57†L226-L229】 . Supplements often provide 15–30 mg/day (elemental zinc). Taking more than 40 mg/day chronically can deplete copper and cause anemia 【60†L24-L29】 . Therefore, most people shouldn't exceed ~40 mg/day (the UL).
- **Timing:** Can be taken with a meal to avoid stomach upset (though food can slightly reduce absorption). Avoid taking zinc at the same time as iron (>25 mg) or calcium, as they compete; space these minerals by a couple of hours 【57†L322-L330】 .
- **Benefits:** Zinc's **evidence strength** is strongest for treating deficiency. If deficient, zinc supplementation restores immune defenses and wound healing. In healthy people, zinc lozenges ( $\geq 75$  mg/day) can slightly shorten cold duration (by ~2 days) 【60†L25-L29】 . Athletically, maintaining zinc helps maintain testosterone and immune health, but studies show minimal performance gains from zinc unless correcting low levels.
- **Side Effects:** Excess zinc causes nausea, vomiting, and can impair immune function due to copper deficiency 【60†L24-L29】 . Long-term high intake (>100 mg/day) may lower HDL cholesterol 【60†L24-L29】 . Overuse of nasal zinc has been linked to loss of smell (anosmia).
- **Age:** Children: RDA increases with age (e.g. teens 9–11 mg). Pregnant and lactating women need ~12–13 mg. The elderly may absorb zinc poorly and often have marginal intake, so supplementation is common. Never give infants large amounts unless prescribed.
- **Contraindications:** People with copper deficiency or on penicillamine (for Wilson's disease) should avoid extra zinc. Caution with certain antibiotics: tetracyclines/fluoroquinolones form complexes (take zinc 2–4 h apart).
- **Target Audience:** Those at risk of deficiency: vegetarians (phytates in plants inhibit zinc absorption), people with GI diseases (Crohn's, ulcerative colitis), alcoholics, the elderly, pregnant/breastfeeding women, and smokers. Also, zinc can be part of a "post-exercise immunity" strategy, but evidence in athletes is weak.
- **Quality:** Zinc comes as sulfate, acetate, gluconate, citrate, etc. Different forms have slightly different absorption 【57†L322-L330】 . Chelated forms (e.g. zinc bisglycinate) may cause fewer GI issues. Choose third-party-tested multivitamins or single supplements to ensure dose and purity.
- **Cost-Benefit:** Zinc supplements are cheap. Given the risk of deficiency in certain groups, moderate supplementation is often justified. However, taking a high-dose zinc supplement "just in case" when diet is adequate is not strongly supported, since excess offers no benefit.

## Vitamin B12

- **Mechanism:** B12 (cobalamin) is required for DNA synthesis, red blood cell formation, and nervous system health (myelin formation). It's a coenzyme for methionine synthase and methylmalonyl-CoA mutase, critical in metabolism.
- **Dosage:** RDA is 2.4  $\mu\text{g}$ /day for adults 【62†L239-L245】 . Supplements typically provide 50–500  $\mu\text{g}$  (as cyanocobalamin or methylcobalamin) because only a fraction is absorbed. Higher oral doses (1000  $\mu\text{g}$ ) are often used in deficiency to ensure enough intrinsic factor-mediated absorption.

- **Timing:** Can be taken at any time, with or without food. Because absorption relies on intrinsic factor and the ileum, splitting doses or taking a large single dose (1000 µg) has been shown to normalize levels in deficiency as well as injections 【64†L103-L112】 .
- **Benefits:** Strong evidence exists that correcting B12 deficiency prevents anemia and neurological damage 【64†L60-L64】 . In people with malabsorption (e.g. pernicious anemia, gastric surgery), high-dose oral B12 can be as effective as injections 【64†L103-L112】 . However, extra B12 has not shown performance or energy enhancement in non-deficient individuals.
- **Side Effects:** Vitamin B12 is water-soluble and non-toxic even at high doses 【63†L7-L10】 . No tolerable upper limit is set because toxicity is extremely rare. Occasionally, high-dose supplements cause minor GI upset or headaches, but generally very safe.
- **Age:** Babies need about 0.4–0.5 µg (via breastmilk or formula), children's RDA rises with age (1–1.8 µg) 【62†L239-L245】 . Adolescents and adults need 2.4 µg; pregnant/lactating women need slightly more (2.6–2.8 µg). Seniors may have trouble absorbing B12 from food (due to low stomach acid); many clinicians recommend B12 supplements for those >60.
- **Contraindications:** None known in general; vitamin B12 is safe even in pregnancy and breastfeeding. Those with folate deficiency may mask anemia but not neurological effects (so both should be monitored together). Rarely, people with Leber's disease (an optic nerve condition) may experience harm from B12 (avoid supplement).
- **Target Audience:** Strict vegetarians/vegans (no natural B12 in plants); older adults and anyone with gastric bypass, pernicious anemia, celiac/Crohn's disease, or on long-term metformin/PPIs (these drugs reduce B12 absorption). Methylcobalamin form may be favored for those with kidney issues or cyanide sensitivity, otherwise cyanocobalamin is stable and effective.
- **Quality:** Choose methylcobalamin if on a B-complex or if concerned about cyanide moiety (especially in smokers/kidney disease). Many multivitamins contain 2–25 µg, but stand-alone B12 often provides 500–1000 µg. Sublingual forms are available, though oral tablets also work well since a portion is absorbed. Look for USP or NSF certification.
- **Cost-Benefit:** B12 is extremely inexpensive. Given the prevalence of subtle deficiency and its importance, supplementing a small dose is a low-cost insurance. High-dose formulations (500–1000 µg) cost more but can correct deficiency.

## Iron

- **Mechanism:** Iron is a key component of hemoglobin in red blood cells and myoglobin in muscle – essential for oxygen transport. It is also involved in energy enzymes (e.g. cytochromes). Iron balance affects endurance and overall energy levels.
- **Dosage:** RDA for adult men is 8 mg; women of childbearing age is 18 mg (due to menstrual losses); pregnant women 27 mg 【57†L226-L229】 . Supplements typically come as 18–65 mg ferrous iron. To treat deficiency (anemia), 60–120 mg elemental iron per day (usually ferrous sulfate, gluconate, or fumarate) is used, divided into 2–3 doses. New research suggests alternate-day dosing may improve absorption and reduce side effects (e.g. 60 mg every other day) – consult current literature.
- **Timing:** Iron is best absorbed on an empty stomach, but can cause nausea. Taking it with vitamin C (orange juice) enhances absorption; avoid calcium or high-fiber foods at the same time. If upset, take with a small amount of food (but no milk/coffee/tea around dosing).
- **Benefits:** **Strong** evidence that iron corrects iron-deficiency anemia, improving hemoglobin and energy levels. In athletes with low iron stores, supplementation can improve endurance and reduce fatigue. However, iron pills do not enhance performance in iron-replete individuals.

- **Side Effects:** Common side effect is gastrointestinal: constipation or diarrhea and stomach upset. Taking the dose in the evening can sometimes mitigate nausea. Overdose is dangerous: acute iron poisoning (mostly in children) can cause organ failure. The UL for adults is 45 mg/day (supps) due to toxicity risk. Chronic excess (hemochromatosis) can cause organ damage (liver, heart, pancreas).
- **Age:** Infants >6 months need iron-fortified formula or cereals (breastmilk is low in iron). Children 1–13: RDA 7–10 mg. Teenage girls 14–18: 15 mg; boys 11 mg [57†L226-L229] . Menstruating women 18 mg. Senior men/women >50: 8 mg (post-menopause women have lower needs).
- **Contraindications:** Do **not** supplement without deficiency in populations at risk of iron overload (e.g. hereditary hemochromatosis, sickle cell, thalassemia). In infection or inflammation, iron supplementation is often withheld because bacteria use iron (consult physician). People with kidney disease or gastrointestinal ulceration should use caution.
- **Target Audience:** Primarily women of reproductive age, pregnant women, infants/young children (especially low birthweight), and people diagnosed with iron-deficiency anemia. Also long-distance runners or endurance athletes (especially females) should monitor iron stores.
- **Quality:** Use ferrous forms for better absorption. Most prescriptions are ferrous sulfate (20% elemental iron), but gluconate/fumarate (higher proportion elemental) may cause less GI upset. Look for non-enteric coated products (slow-release forms are less well absorbed). Avoid multivitamins with iron unless needed, as excess iron can interact with other nutrients.
- **Cost-Benefit:** Iron tablets are very cheap. Given the risk and prevalence of iron-deficiency anemia (especially in women), supplementation when indicated is cost-effective. Unnecessary iron in those without deficiency provides no benefit and potential harm.

## Multivitamin/Mineral

- **Mechanism:** Multivitamins contain a broad array of vitamins/minerals at levels near the daily recommended values. The idea is to prevent nutrient gaps in the diet. There is no single “mechanism” beyond covering multiple micronutrient needs.
- **Dosage:** Typically 1 tablet or capsule daily, providing ~100% of most DVs (Daily Values) for common vitamins and minerals. Dose depends on formulation (e.g. extra B-vitamins or minerals for specific groups).
- **Timing:** Take with a meal (to improve absorption, especially of fat-soluble vitamins). Separate from other supplements/minerals to avoid interactions (e.g. take iron-containing multis away from calcium or magnesium supplements).
- **Benefits:** For generally healthy, well-fed adults, evidence of benefit is *weak*. Large cohort studies find no significant reductions in mortality or heart disease from daily multivitamin use [71†L340-L347] . Some meta-analyses suggest minor reductions in cancer risk for specific groups (e.g. men) but data are inconsistent. Multivitamins **do** prevent deficiencies in people whose diet is poor or who have higher needs (e.g. pregnant women, elderly, those on restricted diets). They ensure a safety net for marginal diets.
- **Side Effects:** Overdoing fat-soluble vitamins can cause toxicity (e.g. vitamin A or D), though most multis limit these. High-dose minerals (like iron or vitamin E) can be problematic for certain groups. Niacin, B6 or excessive vitamin C can cause flushed skin or GI upset at high doses. Generally, common adult multis are considered safe for long-term use in healthy adults.
- **Age:** Children need age-appropriate formulations (chewables or liquids with lower doses). Pregnant women take prenatal vitamins (extra folate, iron). Seniors sometimes need more B12 or D, which some multivitamins provide. Those on medically prescribed diets (e.g. weight loss, food allergies) may use multivitamins if their diet lacks variety.

- **Contraindications:** Smokers should avoid high-dose beta-carotene (vitamin A) due to lung cancer risk. People with hemochromatosis should avoid iron-containing multis. Those with kidney stones history might limit certain forms of vitamin C or calcium. Always check with a doctor if on medication (e.g. vitamin K can interfere with warfarin).
- **Target Audience:** Individuals with **poor dietary intake** (e.g. picky eaters, low fruit/veg intake, elderly living alone), women of childbearing age (for folate and iron), pregnant (prenatal vitamins), vegans (B12, D, maybe iron), and anyone with medically restricted diet. Athletes with very high calorie needs might use multis to cover nutrient loss, though a balanced diet is preferred.
- **Quality:** Look for brands with USP, NSF, or ConsumerLab seals. Avoid “megadoses” unless directed by a doctor. Ensure it contains nutrients suited to your needs (e.g. some older adults need extra B12 or D). Non-essential additives (“proprietary blends”) are unnecessary.
- **Cost-Benefit:** Multivitamins are inexpensive, but their benefit for healthy adults with good diets is questionable [71†L340-L347] . They can be seen as insurance for nutrient gaps. The cost may be worth it if diet is poor, but for those eating varied foods, a multivitamin offers minimal proven benefit.

## Beta-Alanine

- **Mechanism:** Beta-alanine is a precursor to carnosine in muscle. Carnosine acts as an acid buffer, delaying muscle fatigue during high-intensity exercise. By raising muscle carnosine levels, beta-alanine can improve performance in activities lasting about 1–4 minutes.
- **Dosage:** Research suggests **4–6 g/day** of beta-alanine (often split into smaller doses to avoid tingling) for at least 2–4 weeks [75†L2-L4] . This loading period increases muscle carnosine stores by ~60–80%.
- **Timing:** Doesn’t need to be around workouts; daily timing is flexible. Taking doses throughout the day (e.g. 1–2 g four times) minimizes paresthesia (tingling). Some combine with training, but acute timing isn’t critical.
- **Benefits:** Evidence is **moderate**. Meta-analyses show small to moderate improvements (~1–4% performance gain) in high-intensity exercise capacity (like repeated sprints, 400–800m running) [75†L2-L4] . It may also reduce muscle fatigue and soreness slightly. It is less effective for pure strength or endurance >10 minutes.
- **Side Effects:** The only notable side effect is paresthesia (a harmless tingling or flushing sensation), especially with single doses >800 mg. It can be mitigated by time-release formulas or splitting doses. Beta-alanine appears safe otherwise, with no serious effects reported at recommended doses.
- **Age:** Studies are in adults; no data for children. Probably best reserved for adults/late teens in training.
- **Contraindications:** No known medical contraindications in healthy people. Pregnant/nursing women should avoid or use only if directed (no pregnancy safety data).
- **Target Audience:** Sprint athletes, rowers, swimmers, and team-sport athletes who engage in high-intensity intermittent efforts may benefit from beta-alanine. It’s not needed for long-distance or low-intensity sports. Vegetarians and vegans might consider it too, since diet provides only ~1 g of beta-alanine per day.
- **Quality:** Only supplement with pure beta-alanine (no blends with caffeine for instance). Third-party tested products exist but fewer as it’s unregulated; verify label claims.
- **Cost-Benefit:** Beta-alanine products are moderately priced. Given the limited (though positive) evidence, it’s a modestly beneficial supplement for the right sports. Athletes should weigh its small performance gains against cost.

## Caffeine

- **Mechanism:** Caffeine is an adenosine receptor antagonist in the brain, producing alertness and reducing perceived effort. It also raises adrenaline, mobilizes fat, and can improve neuromuscular function.
- **Dosage:** The ergogenic dose is about **3–6 mg/kg body weight** (roughly 200–400 mg for a 70–80 kg person) [84†L65-L74] . Even as low as 2 mg/kg may help. Doses above 9 mg/kg offer no further benefit and greatly increase side effects [84†L65-L74] . 400 mg/day is a common safe upper limit for healthy adults.
- **Timing:** Peak blood caffeine is ~30–90 min after ingestion. Common practice is 30–60 min before exercise [84†L69-L74] . Effects can last 3–5 hours, so avoid late-day doses to prevent sleep disruption.
- **Benefits: Strong** evidence that caffeine acutely improves many aspects of performance [84†L65-L74] . It enhances endurance (running, cycling), muscular strength/endurance, sprinting, and even skill-based tasks. ISSN notes moderate-to-large benefits in aerobic events [84†L61-L69] . It also reliably sharpens focus and reaction time. Benefits occur in both trained and untrained individuals [84†L73-L80] .
- **Side Effects:** Common: jitteriness, anxiety, rapid heart rate, stomach upset, and impaired sleep (especially in sensitive individuals or at higher doses). Tolerance can develop: habitual users may need more for the same effect. High doses ( $\geq 9$  mg/kg) dramatically increase risk of anxiety, GI issues, heart palpitations [84†L65-L74] .
- **Interactions:** Caffeine can interact with certain medications (e.g. some antibiotics, theophylline, MAO inhibitors). It can also worsen side effects of stimulants or counteract sedatives.
- **Age:** Generally safe for adolescents in low to moderate doses (~100–200 mg) with supervision, but children should avoid stimulants. Most research is in adults.
- **Contraindications:** People with uncontrolled hypertension, arrhythmias, or anxiety disorders should use low doses or avoid. Pregnancy limit is often recommended  $<200$  mg/day (since high caffeine may affect fetal growth).
- **Target Audience:** Almost any exerciser can use caffeine for a performance boost or alertness. It is especially favored by endurance athletes and those needing a mental edge (e.g. late training, competitions). Night-shift workers or jetlagged individuals use it to counter fatigue.
- **Quality:** Sources include coffee, tea, chocolate, or supplements (pill/capsule, energy drinks). For sports, anhydrous caffeine pills are precise. Third-party testing (NSF, Informed-Choice) is wise, especially for athletes subject to doping rules (caffeine is not banned but products should be pure).
- **Cost-Benefit:** Caffeine is very cheap and effective. It's one of the most reliable legal "performance enhancers." For most healthy adults, the benefits of moderate caffeine use outweigh the cost (cost is negligible). However, one must self-regulate dose and timing to avoid negative effects.

## Citrulline Malate

- **Mechanism:** Citrulline is converted to arginine in the kidneys, boosting nitric oxide (NO) production, which dilates blood vessels and may improve muscle blood flow and nutrient delivery. It also helps clear ammonia by cycling through the urea cycle, potentially delaying fatigue.
- **Dosage:** Studies commonly use **6–8 g of citrulline malate** (often 2:1 ratio, meaning 4–5 g citrulline + 2–3 g malate) taken about 1 hour before exercise.
- **Timing:** Take on an empty stomach 45–60 min pre-workout to allow conversion to arginine. Dividing smaller doses is also possible, but one large dose is typical for performance.

- **Benefits:** Evidence is **limited to modest**. Some trials show improved repetitions to fatigue and reduced muscle soreness with ~8 g 【87†L1-L4】. Meta-analyses are mixed, but many users report better “pump” or endurance during moderate-length efforts. Performance gains, if any, tend to be small.
- **Side Effects:** Generally safe. Very high doses (>10 g) can cause stomach discomfort. No serious issues reported at standard doses.
- **Age:** Used in adults only; no data for children or teens.
- **Contraindications:** None known; avoid if you have low blood pressure (vasodilatory effect). Those on nitrates or ED medications (which also increase NO) should be cautious to avoid hypotension.
- **Target Audience:** Mainly weightlifters or athletes doing intense circuit training or anaerobic repeats (Citrulline may help with muscle “pump” and delay fatigue). It’s less useful for brief sprints or long cardio.
- **Quality:** Look for **citrulline malate** specifically (some products use free citrulline or L-arginine, which are less effective). The ratio (2:1) should be on the label; ensure it’s L-citrulline malate (not filler). Prefer supplement tested for actual dose.
- **Cost-Benefit:** Citrulline malate supplements are moderately priced. Given the uncertain evidence, they are not “must-haves,” but for those curious about NO boost, a trial may be reasonable. Athletes should weigh anecdotal benefits against cost.

## Ashwagandha

- **Mechanism:** Ashwagandha (*Withania somnifera*) is an adaptogenic herb. It modulates the stress response (may lower cortisol), and some studies suggest it can boost testosterone and muscle strength via the hypothalamic-pituitary-gonadal axis 【93†L1-L4】. It also has GABAergic (calming) properties.
- **Dosage:** Standardized root extract is usually 300–600 mg once or twice daily (total 600–1200 mg/day) of a preparation containing standardized withanolides (5–10%). Sensoril and KSM-66 are common brands.
- **Timing:** Can be taken with meals (to avoid stomach upset). Evening dosing may aid sleep. No precise “timing” relative to workouts is needed.
- **Benefits:** Clinical trials and meta-analyses show **moderate** benefits in reducing stress and anxiety, improving sleep quality, and lowering cortisol levels. Some RCTs report modest increases in muscle mass/strength and testosterone in men taking ashwagandha (300–600 mg) for 8+ weeks 【93†L1-L4】. It may slightly enhance VO<sub>2</sub>max and cognitive function under stress.
- **Side Effects:** Generally well-tolerated. Occasional stomach upset or diarrhea can occur. Large doses or extracts with high withanolide content may cause sedation. Traditional Ayurvedic sources advise that it can stimulate uterine contractions, so it is *not recommended in pregnancy or breastfeeding*.
- **Age:** Studies are in adults. Should not be given to young children without practitioner guidance.
- **Contraindications:** Avoid if pregnant or breastfeeding. People on thyroid medication should use caution (ashwagandha can affect thyroid hormones). Always check for herb–drug interactions; it may potentiate sedatives or immunosuppressants.
- **Target Audience:** Stressed or anxious adults, athletes in heavy training (to mitigate stress), men with low-normal testosterone, and older adults concerned with vitality. Also used to support sleep when stress is high.
- **Quality:** Look for standardized extracts (withanolide content listed) and reputable brands. KSM-66 and Sensoril are common clinical extracts. Organic certification can reduce pesticide concerns. Third-party testing is a plus.

- **Cost-Benefit:** Ashwagandha is moderately priced. Given its stress-reduction benefits (supported by several trials), many find it worthwhile. The performance benefits (strength/testosterone) are smaller and might not justify cost alone.

## Collagen (Hydrolyzed and UC-II)

- **Mechanism:** Collagen supplements provide amino acids (glycine, proline, hydroxyproline) needed to build connective tissue. Hydrolyzed collagen (collagen peptides) is digested into small peptides that may stimulate skin and cartilage repair. Undenatured Type II collagen (UC-II) works via immune tolerance, which can help with cartilage health in joints.
- **Dosage:** Hydrolyzed collagen is used at **2.5–15 g/day** (common dose ~10 g). For undenatured collagen (UC-II), the effective dose is much lower (~40 mg/day) **【95†L280-L286】** .
- **Timing:** Any time of day; taking with vitamin C can aid collagen synthesis (C is a cofactor for collagen cross-linking). Even division (morning vs evening) is fine.
- **Benefits:** Evidence is **moderate**. For skin, multiple trials show that 1–10 g/day of collagen peptides improves skin hydration, elasticity, and reduces wrinkles after ~8–12 weeks **【95†L241-L249】** . For joints (osteoarthritis), collagen studies (peptides or UC-II) report reduced pain/stiffness and improved function **【95†L254-L263】** . A meta-analysis of 25 trials found collagen “benefited OA and aided cartilage repair” regardless of type **【95†L262-L264】** .
- **Side Effects:** Collagen is well-tolerated. Rare complaints include mild GI upset or fishy aftertaste (if fish-derived). Because it's a protein, those with severe protein allergies should avoid related sources (e.g. some with pork allergy avoid bovine collagen, though cross-reactivity is uncommon).
- **Age:** Adults (especially >40, when natural collagen declines) benefit most. Not for vegans/vegetarians (no plant collagen sources). Children do produce collagen; supplementation in youth is not typically needed except for injury healing (under a doctor's care).
- **Contraindications:** None major. Those on anticoagulants or with bleeding disorders should note collagen can influence platelet function, so high doses might affect clotting (though evidence is limited). Allergies to source (e.g. fish, egg shell membrane, or beef/pork) are the main concern.
- **Target Audience:** People with joint discomfort (osteoarthritis, overuse injuries) and those wanting skin/beauty benefits. Also used post-injury/rehab to support cartilage repair. Not a typical performance enhancer for athletes, but some weightlifters use it for tendons/joint support.
- **Quality:** Choose **hydrolyzed collagen peptides** (type I & III for skin, type II for joints) for easy mixability, or UC-II specifically for OA support. Look for reputable sources (bovine, porcine, marine) and check for heavy metal testing (fish collagen can concentrate mercury if source is poor). Third-party tested gelatin/collagen powders are best.
- **Cost-Benefit:** Collagen supplements are moderately expensive per gram. Given the positive evidence for skin and joint outcomes **【95†L241-L249】** **【95†L262-L264】** , many users find the cost worthwhile for cosmetic or comfort improvements. If budget is a concern, focusing on collagen-rich diet (bone broth, gelatin) may help, though supplement doses are much higher.

## Probiotics

- **Mechanism:** Probiotics are live beneficial bacteria or yeasts that, when ingested in adequate amounts, confer health benefits by modulating the gut microbiome. They can strengthen gut barrier function, modulate immunity, and influence metabolism through the gut–brain–muscle axis **【98†L418-L427】** .

- **Dosage:** Effective doses are strain-specific, typically in the range of  $10^9$ – $10^{11}$  CFU/day. Check label for the species (e.g. *Lactobacillus*, *Bifidobacterium*, *Saccharomyces*). A common single dose is ~1–10 billion CFU in a capsule.
- **Timing:** Often taken on an empty stomach (before breakfast) or with a small meal. Some evidence suggests taking with food increases survival through the stomach. Consistency is key (daily use for at least several weeks).
- **Benefits:** Probiotic benefits vary by strain and population. In healthy athletes, probiotics appear to provide *modest* improvements: a 2026 meta-analysis found a small-to-moderate boost in overall athletic performance, especially endurance ( $VO_2\max$ ) [98†L362-L371] . Many trials report reduced incidence/duration of upper-respiratory and gut infections in athletes. For the general population, benefits include improved gut health (e.g. less diarrhea after antibiotics) and enhanced immune response. Certain strains (e.g. *Lactiplantibacillus plantarum*) are highlighted for endurance gains [98†L368-L372] .
- **Side Effects:** Generally safe for healthy people. Possible transient bloating or gas for a few days. In immunocompromised or critically ill patients, rare cases of bacteremia or fungemia have occurred, so those individuals should only take probiotics under medical advice.
- **Age:** Special formulations exist for infants and children (e.g. *Bifidobacterium infantis* in baby drops). Adults and seniors also use probiotics. Dosage and strains may differ by age/condition.
- **Contraindications:** Weakened immune systems (HIV/AIDS, chemotherapy) or severe pancreatitis, central line presence, recent surgery are relative contraindications (due to infection risk). Those on live vaccines should separate timing by a few weeks (theoretical concern).
- **Target Audience:** Anyone with gut or immune concerns: travelers, antibiotic users, IBS sufferers (certain strains help diarrhea/constipation), and athletes (to reduce GI distress and boost immunity). Probiotics are not a direct “performance supplement” like creatine, but good gut health can indirectly support training consistency.
- **Quality:** Use products specifying genus, species, strain (e.g. *L. rhamnosus* GG, *B. animalis*). Look for clinically studied strains. Check CFU count on the label and expiration date (viability declines over time). Third-party tests (e.g. NSF, USP, ConsumerLab) ensure the CFUs listed are accurate and contaminants are absent.
- **Cost-Benefit:** Quality probiotics can be expensive. Given the potential benefits for gut and immune health (and even minor performance gains) [98†L362-L371] , they may be worth it for those with gut issues or heavy training loads. If one’s diet includes fermented foods (yogurt, kefir, sauerkraut), some benefit is obtained. Supplements are most beneficial when targeted to a specific need (e.g. preventing traveler’s diarrhea, or reducing URTIs in athletes).

## Electrolytes

- **Mechanism:** Electrolytes (sodium, potassium, magnesium, calcium, etc.) maintain fluid balance, nerve conduction, and muscle function. During exercise, sweating leads to electrolyte loss, which can cause cramps, fatigue, or hyponatremia if fluids are replaced without salts.
- **Dosage:** There’s no “daily supplement” dose; instead, electrolytes should replace what’s lost in sweat. A rough guideline: for intense exercise, sports drinks often contain ~300–700 mg sodium and 80–200 mg potassium per liter. Magnesium and calcium losses are smaller (tens of mg). Specific needs vary widely.
- **Timing:** Electrolytes should be consumed **during or after** prolonged exercise. Pre-loading (e.g. a salty meal before a marathon) can help. For regular daily needs, a balanced diet usually covers it.

- **Benefits:** Replacing electrolytes maintains plasma volume and prevents dilutional hyponatremia during long workouts. It can reduce the risk of muscle cramps (especially if sodium loss is high) and improve endurance by sustaining nerve and muscle function.
- **Side Effects:** Unlikely at normal replacement doses. Excess sodium in non-sweating individuals can elevate blood pressure. Oral potassium in high doses can upset the stomach. Overall, balance is key.
- **Age:** Children can sweat heavily, but their electrolyte needs scale with body size. Kids doing long-duration sports should use diluted sports drinks or electrolyte solutions. Elderly or sedentary people typically only need dietary electrolytes unless unwell.
- **Contraindications:** People with kidney disease or heart conditions should monitor electrolyte intake (especially potassium). Those on ACE inhibitors/ARBs or potassium-sparing diuretics shouldn't take high potassium loads.
- **Target Audience:** Endurance athletes, team-sport players, or workers in hot environments who sweat a lot. Also anyone with salt loss (e.g. through diarrhea/vomiting).
- **Quality:** Sports drinks, electrolyte tablets, or powders usually contain balanced mixtures of Na, K, Mg, Ca. Check labels to avoid excessive sugar or unneeded additives. NSF Informed-Sport certifies some electrolyte products for athletes.
- **Cost-Benefit:** Electrolyte supplements are moderately priced. For someone training intensely, the benefit of preventing dehydration and cramps is clear. Casual exercisers on a normal diet rarely need extra electrolytes beyond water unless they sweat profusely.

## Glutamine

- **Mechanism:** Glutamine is a conditionally essential amino acid. It's a key fuel for intestinal cells and immune cells, and a nitrogen transporter between tissues. During severe stress (burns, trauma), glutamine demand exceeds supply.
- **Dosage:** Common supplement doses are 5–10 g/day. It can be taken once or split (e.g. before and after exercise).
- **Timing:** Often taken post-exercise in recovery shakes. It is also marketed for gut support (on an empty stomach for intestinal cells).
- **Benefits:** For healthy athletes, **evidence is weak**. Glutamine does not significantly enhance muscle growth or performance when dietary protein is adequate. Some suggest it may help immune function or gut integrity after extreme exercise, but this is mostly seen in clinical (ICU) settings or severe stress. In athletes, any recovery benefit is minimal.
- **Side Effects:** Generally regarded as safe. Doses up to 20–30 g/day have been used with few issues, but large amounts can cause stomach upset or affect ammonia levels.
- **Age:** Studied mostly in adults. It's naturally abundant in foods (meat, dairy); children synthesize it adequately. Supplemental use is not needed in normal pediatrics.
- **Contraindications:** Generally none. People with liver or kidney disease should not take large doses (impaired ammonia metabolism).
- **Target Audience:** Might be considered by athletes undergoing heavy training or high catabolic stress (though evidence is low). Also used in clinical nutrition (hospital patients). For typical gym-goers, it's not necessary if protein intake is sufficient.
- **Quality:** Look for L-glutamine powder (avoid Greek yogurt mixes claiming high glutamine, since freezing yogurt doesn't concentrate it). Check for purity.
- **Cost-Benefit:** Glutamine is inexpensive. But given the lack of clear benefit for performance/recovery, it's not high priority. It may be more useful in wound healing or gut health in special cases.

## BCAAs (Leucine, Isoleucine, Valine) / EAAs

- **Mechanism:** BCAAs (especially leucine) directly stimulate muscle protein synthesis, bypassing the liver. EAAs provide all essential amino acids needed for MPS. The idea is to reduce muscle breakdown and speed recovery.
- **Dosage:** Common BCAA supplements contain 5–10 g total (often 2:1:1 ratio of leucine:isoleucine:valine). EAA supplements might provide 5–15 g (including 2–3 g leucine).
- **Timing:** Taken around workouts or between meals to potentially reduce soreness. Some drink them during workouts on an empty stomach.
- **Benefits:** If you consume adequate complete protein daily, BCAAs alone add little. BCAAs may slightly reduce muscle soreness, but evidence that they improve performance or muscle gains beyond what dietary protein does is **weak**. EAAs (full spectrum) are more effective than BCAAs for stimulating MPS, but again only if overall protein is low. Athletes on very low-protein diets might see some benefit; otherwise, whole protein (whey/EAA) is superior.
- **Side Effects:** High intake of leucine alone can cause temporary fatigue or interfere with other amino acids. Pure BCAAs can upset blood sugar control if taken without food. No serious toxicity at typical doses.
- **Age:** Adults. Children get EAAs from diet; no need to supplement.
- **Contraindications:** Those with certain metabolic disorders (e.g. MSUD) should avoid BCAAs. Otherwise none major.
- **Target Audience:** People eating very little protein (e.g. on fasting or very low-cal diets) sometimes use BCAAs to preserve muscle. They are not needed if hitting protein goals with diet or whey/EAA supplements. Some vegans use them in addition to plant protein to boost leucine intake.
- **Quality:** BCAAs should list the ratio and mg of leucine. EAAs should list all nine EAAs. Verify there's no proprietary blend hiding actual amounts.
- **Cost-Benefit:** BCAAs and EAAs are relatively expensive per gram of active ingredients. Given their minimal advantage over whole protein, they are often low on the priority list for nutrient spending. If budget allows, whey/EAA products are generally more cost-effective and evidence-backed for supporting muscle.

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**Sources:** Authoritative reviews, meta-analyses and position stands from sports nutrition societies and governmental health organizations have been used to compile these recommendations [7†L79-L87] [10†L268-L277] [20†L833-L841] [36†L94-L102] [38†L376-L382] [40†L33-L39] [44†L251-L254] [50†L148-L157] [52†L703-L710] [60†L24-L29] [64†L60-L64] [71†L340-L347] [84†L65-L74] [98†L362-L371] . These represent current consensus and research as of 2026. Always read labels (remove tracking tags) and prefer third-party tested products (NSF, Informed-Sport, USP).

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